

B.Ranhitha
2403a51247
Batch—11

Set-F

F–G: TDD with AI

F.1 — [S18F1] Add type hints and fix None handling

Context:

A helper in real estate listings platform must never return None silently.

Your Task:

Annotate and raise ValueError if target missing.

Data & Edge Cases:

ids=['a','b'], target='c' -> ValueError('not found').

AI Assistance Expectation:

Use AI to propose mypy-friendly signature.

Constraints & Notes:

Add negative test for miss.

Sample Input

ids=['a','b'], target='c'

Sample Output

ValueError('not found')

Acceptance Criteria: Raises correctly; typed

prompt: Write a Python program to search for a target in a list of ids. Return the target if found, otherwise raise `ValueError("not found")`. Use type hints and test both found and not-found cases.

Code:

```
[1]
✓ 0s

from typing import List

def find_id(ids: List[str], target: str) -> str:
    """
    Finds the target in the list of ids.
    Raises ValueError if target is missing.
    """
    if target in ids:
        return target
    raise ValueError("not found")

# ✓ Positive test
print(find_id(['a', 'b'], 'a')) # Expected: 'a'

# ✗ Negative test
try:
    print(find_id(['a', 'b'], 'c'))
except ValueError as e:
    print("Error:", e)
```

Output:

```
↵ a
Error: not found
```

F.2 — [S18F2] Optimize membership checks

Context:

Streaming membership checks in real estate listings platform.

Your Task:

Convert corpus to set and map stream to booleans.

Data & Edge Cases:

corpus=[1,2,3,4,5], stream=[2,5,9] -> [True, True, False].

AI Assistance Expectation:

Ask AI for complexity analysis and tests.

Constraints & Notes:

Return aligned list[bool].

Sample Input

corpus=[1,2,3,4,5], stream=[2,5,9]

Sample Output

[True, True, False]

Acceptance Criteria: Uses set; correct

Prompt: Write a Python program to check membership of stream elements in a corpus by converting the corpus to a set. Return a list of booleans aligned with the stream, and test with edge cases.

Code:

```
[2]
✓ 0s
from typing import List

def membership_check(corpus: List[int], stream: List[int]) -> List[bool]:
    """
    Converts corpus to a set and maps each stream element
    to True/False depending on membership.
    """
    corpus_set = set(corpus) # O(n)
    return [item in corpus_set for item in stream] # O(m)

# ✓ Test case 1
print(membership_check([1, 2, 3, 4, 5], [2, 5, 9]))
# Expected: [True, True, False]

# ✓ Test case 2 (edge: empty corpus)
print(membership_check([], [1, 2]))
# Expected: [False, False]

# ✓ Test case 3 (edge: empty stream)
print(membership_check([1, 2, 3], []))
# Expected: []
```

Output:

```
⇒ [True, True, False]  
  [False, False]  
  []
```